Amey Kulkarni

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Education

Pune Institute of Computer Technology (SPPU) | *B.E. Computer Engineering, CGPA 9.45/10* 2022 – 2026

Experience

Espressif Systems Pune, India

Associate Software Engineer (converted from internship) *Jul 2026 – Present*

- Own **ESP User**, Espressif's open-source identity service; rebuilt it into an **OAuth 2.0 / OpenID Connect authorization server** in Go issuing audience-scoped JWTs (**RFC 8707/9068**) valid for one API only
- Delivered **single sign-on across Espressif products**: device-bound sessions, session-bound refresh-token families and RP-initiated logout let users sign in once, list their devices and sign out everywhere
- Added the **client_credentials** grant for service-to-service auth, unifying token issuers on one signing key
- Implemented the **Device Authorization Grant (RFC 8628)** for SSO sign-in on headless and screen IoT devices
- Designed **migration-safe identity linking** that keeps product data intact as users move behind ESP User

Cloud Backend Intern *Jan 2026 – May 2026*

- Architected **ESP Cloud Hub** (shipped **v1.0.0**), a **multi-tenant control plane** deploying Espressif products into customers' own AWS accounts via **cross-account STS AssumeRole** with HMAC-derived ExternalIds
- Enforced **tenant isolation** with an **ABAC** policy engine (role catalogue, deny-override, org-scoped grants) on API Gateway, Lambda, DynamoDB and Cognito; every request is scoped to its organisation and fails closed
- Built **SSO into Cloud Hub**: OAuth 2.0 + PKCE federation and **AWS IAM Identity Center** device-authorization tenant connect, with session tokens kept server-side in SSM
- Engineered a **DAG deployment orchestrator** on **AWS Step Functions** (Kahn's algorithm) that runs independent CloudFormation stacks in parallel waves with cycle detection and resume-from-failure

Mastercard Pune, India

SDE Intern, Mastercard Checkout Services *May 2025 – Jul 2025*

- Built **Karate** test automation for production **Spring Boot / Spring Security** checkout services, covering untested flows, plus end-to-end tests for the notification email service, run in **Jenkins CI**
- **Top 5 of 30** teams, Global Innovation Challenge: AI concept for restaurant SMEs (11 interns, 4 countries)

Technohertz Pune, India

Application Developer Intern *Sep 2024 – Dec 2024*

- Built a Skoda employee social app (**Flutter**, **Django** REST, FCM) and a **Python RAG chatbot** for employee queries

Projects

NeuroDB: Self-Adaptive Distributed Database | *B.E. capstone; FastAPI, PostgreSQL* 2025 – 2026

- Built a sharded, replicated SQL layer over **PostgreSQL**: catalog service, a router that **hash-partitions** rows onto **virtual shards**, primary–replica nodes, and a **rebalancer** that re-shards as nodes join or leave
- Added **ML-driven join ordering** for query optimisation and **anomaly detection** on node telemetry that migrates data before a predicted node failure

Multilingual Serverless Video Search | *AWS Lambda, API Gateway, DynamoDB, BGE-M3* 2026

- Semantic search over Marathi discourses that returns the exact video timestamp for English or Marathi queries
- **BGE-M3** multilingual embeddings for cross-lingual retrieval; pay-per-request **API Gateway + Lambda** backend

Technical Skills

Languages: Go, Python, C++, SQL (PostgreSQL)
Identity & Security: OAuth 2.0, OpenID Connect, SSO, PKCE, Device Authorization Grant, JWT, Cognito, ABAC
AWS: Lambda, API Gateway, Step Functions, DynamoDB, S3, STS, IAM, SSM, CloudFormation, SAM, CDK
Frameworks & CI/CD: FastAPI, Django, Flask, React, Flutter, GitLab CI, Jenkins

Achievements & Publications

Hackathons, 7 podiums: 1st Impetus & Concepts 2026 (NeuroDB) & 2025 (Guardian360); 1st Pulzion 2023 & 2024 (QullQuery, QueryCortex); 3rd TechFiesta 2025; 2nd TechFiesta 2024; 3rd Impetus & Concepts 2024
IEEE publications: *Federated Learning for Cardiovascular Disease Prediction* (ESCI 2025); *Hair Loss Stage Classification using Deep Learning* (ICAET 2025)